

An Agentic Approach to a First Price Auction with Complementary Goods

Sam Armstrong

Abstract—In this report I present an agentic approach to a first price painting auction game that uses the Cobb-Douglas product of each bot's painting collection to calculate the winner. This scoring mechanism means agents face an exposure problem where obtaining a balanced set of paintings will yield the highest score, and failing to secure paintings from all artists will yield a score of zero. The proposed bot calculates the true value of paintings based on the expected marginal utility and the relative scarcity of the remaining paintings, which is used as a baseline bid. As first price auctions are not incentive compatible, bid shading is used to help preserve budget, so the bot can win more paintings overall and achieve the highest possible score. The bot is evaluated across multiple room configurations, demonstrating strong performance, and highlighting some shortcomings with the current strategy that could be improved upon in future work.

I. INTRODUCTION

A. The Game

In our first price auction game, a room is set up containing anywhere from two to ten bots, each with a budget of 1001. In each round, the type of painting up for auction will be announced from one of four artists - Picasso, Van Gogh, Rembrandt, or Da Vinci; the bots will then be able to submit a single integer bid and the highest bidder will win the painting, paying the amount of their bid. If the highest bid amount has been bid by multiple bots, the winner is chosen at random, meaning each round only has one winner. The game runs for 200 rounds, and the final score for each bot is determined by the product of the number of paintings they have won from each artist, so if a bot hasn't won any paintings from a given artist, their score will be zero. The bot with the highest score at the end of the game wins. Each bot has access to the order in which the paintings will be auctioned, and the winner and winning bid for each of the previous rounds.

B. Game Theory

1) *Complementarity and Exposure*: The scoring function for this first price auction game is a four-variable Cobb-Douglas function [1] (the product of the count of each variable), which means that the paintings from different artists are complementary. When we have no paintings from a given artist, a painting from that artist is the perfect complement to the other paintings we have, as they are worthless (they give a total score of zero, regardless of how many we have) without it. Due to the complementary nature of the paintings in this game, one of the key considerations for designing strategies is the exposure problem, highlighted by Milgrom (2000), which is 'the phenomenon that a bidder that bids straightforwardly

according to its demand schedule is exposed to the possibility that it may wind up winning a collection of licenses that it does not want at the prices it has bid' [2]. In the context of this game, the exposure problem could manifest as the bot successfully collecting a large number of paintings, but failing to get many from a certain artist, leading to a low overall score. One strategy that can be used to address the exposure problem is bidding based on the expected marginal utility of a painting, which was demonstrated by Greenwald and Boyan (2004) [3] as an optimal bidding policy for sequential auctions with complementary goods. In this game, the expected marginal utility of a painting is dependent on how many paintings we already have from that artist, and how many we have from other artists, as well as how scarce paintings from that artist are in the remaining rounds of the auction.

2) *Budget Constraints*: There are also important considerations to be made around the budget constraints placed on the bots. First, there is an opportunity cost to bots bidding high on paintings - bidding high makes securing a painting more likely, but diminishes their ability to win future paintings. To address this trade-off, budget will need to be logically paced in bids throughout the game, which can be combined with the strategy of bidding based on the expected marginal utility of a painting to create a strategy that bids more on the most important paintings without overspending and risking losing out on important paintings later in the game. The fact that the remaining budget of each bot is known also means that predatory bidding strategies could be used, such as bidding exactly one more than the highest remaining budget of any opponent when they're running low [4].

3) *Incentive Compatibility and Bid Shading*: In this first price auction, the bot has to pay exactly what it bids, meaning bidding the exact valuation of a painting has a net-zero profit. This contrasts with second price auctions (Vickrey auctions [5]), where the bot pays the second highest bid, which are incentive compatible as the bot would be incentivised to bid its true valuation (as this would make it more likely to win and it would still only pay the second highest bid, meaning it would likely make a profit). As our first price auction is not incentive compatible, in order to make a profit (leaving more budget for future paintings), the agent will need to shade its bids down below its true valuation. In a first price auction where the bidders have independent private values drawn uniformly on $[0, 1]$, the bid should be shaded down using a 'Bayesian Nash Equilibrium' approach to updating the expected distribution of opponent bids [6], [7]. However, this doesn't theoretically apply to our auction, where values are determined by Cobb-

Douglas scoring for each bot. Given the information we have available, the shading could instead be calculated based on opponent's inventories and budgets.

II. STRATEGIES

A. Calculating our True Valuation

The first strategy I use for my bot is to calculate our true valuation of the current painting up for auction. My bot does this by first calculating a base bid by dividing the remaining budget by the number of rounds remaining over the number of bots in the room, as with more bots present there will be greater competition for the paintings, so we'll need to bid higher. It then calculates the marginal utility that winning a painting from each artist in the game would give, using a heuristic estimation of how much our score would increase if we won a painting from that artist using the formula $1 + \frac{1}{c+1}$ where c is the number of paintings we have from that artist. This is a good heuristic to use, as the marginal utility of the first painting from an artist is theoretically infinite, because without it the score would be zero. This estimation ensures that our marginal utility is finite and bounded in the range $[1, 2]$, which makes calculating the relative marginal utility score easier and logically represents the diminishing marginal returns of getting more paintings from the same artist. The calculated marginal utility of the current painting is then divided by the average marginal utility over all artists to create a relative marginal utility score for the current painting. This is multiplied by the base bid to get an adjusted bid that reflects the importance of the painting to our current portfolio.

The bot then calculates how scarce paintings from this artist are in the remaining rounds of the auction, relative to how many we'd expect if the paintings were perfectly balanced (the number of rounds remaining divided by the number of artists). This information is important to include in our valuation, as it reflects our capability to secure more paintings from this artist in the future - if this will be difficult it's more important to try to secure this one. This scarcity value is divided by the number of bots in the room and added to one to get a scarcity factor, which is multiplied by the bid to get our true valuation of the painting.

B. Bid Shading

The next strategy used is bid shading, aiming to reduce the bid to a logical amount below our true valuation, which is important to help the bot preserve budget so it can win more paintings overall. The bot first calculates an initial shade that's based on the number of bots in the room. This uses a base Bayesian Nash Equilibrium (BNE) shade, the theoretical optimal bid for a first price auction when the bidders have independent private values, but uses a convex combination to increase the value, so the shade is less aggressive and the bid will remain competitive. This is important in theory, as shading down to the pure BNE fails to consider the interdependent valuations of the paintings, which could lead to underbidding and missing out on critical paintings. The convex combination is calculated using the formula:

$$\text{blended_shade} = b + (1 - b)(1 - b^2)$$

Where b is the BNE shade, calculated using the formula $\frac{n-1}{n}$, n being the number of bots in the room. The $1 - b$ term represents the gap between the BNE shade and 1 (our true valuation), and the $(1 - b^2)$ term represents how much of this gap should be reclaimed to make the bid more competitive. The combination makes the bid more aggressive the more bots there are in the room, but scales it non-linearly to ensure we're always bidding under our true valuation, and within a reasonable range. This also worked better in practice in this game where values are determined by Cobb-Douglas scoring for each bot, improving the win rate in a room against a bot that simply bids its true valuation from 0% with a pure BNE shade to 48% with the adjusted shade. For two player rooms, where the bot is playing against a single opponent, it uses a different custom shade which is calculated using the formula $1 - \frac{1}{n^2(n+1)}$, where n is the number of bots in the room. This produces a smaller shade that was more effective in testing against a single opponent, as budget preservation for winning key paintings in later rounds is less significant in relation to outbidding them in more rounds. This improved the win rate in the room against a bot that bids its true valuation from 48% up to 61.7%.

After calculating the initial shade, the bot checks its recent win rate against its 'fair share' of wins, using a sliding window. If the win rate is under fair share, it calculates an adapt up multiplier, which scales linearly based on how badly we are missing our fair share target, capped at max up. If the win rate is over fair share, it scales down the bid, which works less aggressively than adapt up, as we don't want to reduce our bid too aggressively and make it uncompetitive. Finally, our true valuation is multiplied by the shade to get our final bid, but is not applied in the final rounds so the remaining budget is spent to help secure these paintings, as any remaining budget at the end of the round is wasted.

As I discussed in *Incentive Compatibility and Bid Shading*, using a Bayesian Nash Equilibrium based shade is not theoretically optimal for this where the valuations of a given painting are not independent and private, but I decided to use this approach as we offset these issues using the convex combination, and this method proved to generally yielded the best results in practice compared to other approaches I tested, such as shading based on opponent threat levels, calculated from their marginal utilities and budgets.

C. Applying Budget Constraints

The final strategy used by the bot is applying budget constraints. The bot is capped to never spend more than one over the highest remaining budget of any opponent, as no opponent can ever bid higher than this, so it would be wasted budget. Finally, the bot cannot bid any more than its remaining budget, as this would cause it to bid zero and lose the painting.

D. Alternative Strategies

I tested out using a few alternative strategies that failed to beat the bot I settled on as described above.

1) *Bid Shading Based on Opponent Threat Levels*: One strategy I implemented was estimating the threat level of each opponent by applying the same true valuation logic my bot uses to estimate its own true valuation. I then created a search space to find a shaded bid that maximises the expected profit, given the threat levels of the opponents. The performance of this strategy was varied, only achieving a 29% win rate in a 2 player room against my baseline bot, but winning 61% in a room with the baseline bot and a bot that bids its true valuation. I decided not to use it due to this inconsistent performance, but it did show promise and improve on performance in certain room configurations.

2) *Bid Shading Decay*: I experimented with using a strategy of bid shading decay to make the shade take less affect as the rounds progress, which should theoretically mean the budget is spent more effectively towards the end of the game, when there is little benefit in preserving budget for future rounds. However, this drastically worsened the performance, winning only 72% of games against a bot that always bids 10, so I stuck with my strategy of just removing the shade for the final rounds.

III. RESULTS AND EVALUATION

I evaluated the performance of my bot in several different room configurations, with the results shown in Table I. The bot shows strong performance in all the room configurations I tested, winning a majority of games in most of the rooms. Notably, it has a majority win rate over the bot that bids its true valuation (calculated the same way as my bot) in all rooms it's part of, proving the bid shading strategy is effective at preserving budget, allowing my bot to win more of the most important paintings.

Surprisingly, however, my bot loses around 6% of games to the bot that always bids 10, suggesting there may be room for improvement in the strategy. In contrast, the threat_level bot (discussed in *Alternative Strategies*) achieves around 97.5% win rate in the same configuration, indicating the opponent threat modelling is more effective at bid shading than my existing approach, and could be refined or combined with my existing approach in future work to potentially exceed my bot's performance across the board.

This is further supported by the weak performance (62% win rate) of my bot against the room with 6 flat bots, which should be relatively easy to beat for a bot using advanced strategies. Observing a run of this room shows that the bot is winning a lot of rounds early, bidding significantly more than the flat bots, then losing most of the remaining rounds. It seems that if many of the most important paintings aren't present early, the bot will end up not being able to collect these later on and end up losing. This initially overbidding is likely happening as there are many bots present in the room, so our bot is expecting the bids for paintings to be highly competitive as other bots seek to secure them. Instead, the flat bots are just bidding their standard amount and losing these rounds, but this means our bot quickly ends up depleting its budget.

TABLE I
PERFORMANCE IN DIFFERENT ROOM CONFIGURATIONS

Room	Games	Win Rate (%)
sam_bot, flat_10	300	94
sam_bot, random * 10	100	100
sam_bot, flat_10 * 5	100	100
sam_bot, flat_15 * 5	100	100
sam_bot, flat_10 * 3, flat_15 * 3	100	62
sam_bot, flat_10 * 3, flat_15 * 3, random * 3	100	95
sam_bot, true_value	300	61.7
sam_bot, true_value, flat_15	100	56
sam_bot, true_value, flat_15 * 3	100	100
sam_bot, threat_level	100	71

IV. CONCLUSION

In this report, I discussed the game theory considerations for a first price complementary painting auction game, and presented a bot that implements strategies derived from this theory, achieving strong performance across most tested room configurations, though with some suboptimal performance when playing against multiple flat bots. While the performance is generally good, I also discussed alternative strategies that were tested but not used in the final bot, primarily a bot that implemented bid shading based on opponent threat modelling. This had more varied results than my bot, but showed strong results in some configurations. An approach like this is more theoretically grounded than modified Bayesian Nash Equilibrium based bid shading I used for my bot, as it takes into account the interdependent valuations of the paintings, so it has the potential to be improved upon in future work to potentially exceed my bot's performance. There are also other directions future work could take for an agentic approach to this game, such as using deep reinforcement learning - which has previous been explored and applied to similar auction applications [8].

REFERENCES

- [1] C. W. Cobb and P. H. Douglas, "A theory of production," *The American Economic Review*, vol. 18, no. 1, pp. 139–165, 1928. [Online]. Available: <http://www.jstor.org/stable/1811556>
- [2] P. Milgrom, "Putting auction theory to work: The simultaneous ascending auction," *Journal of Political Economy*, vol. 108, no. 2, pp. 245–272, 2000. [Online]. Available: <http://www.jstor.org/stable/10.1086/262118>
- [3] A. Greenwald and J. Boyan, "Bidding under uncertainty: theory and experiments," in *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence*, ser. UAI '04. Arlington, Virginia, USA: AUAI Press, 2004, p. 209–216.
- [4] M. H. Kotowski, "First-price auctions with budget constraints," Jun. 2019, hKS Working Paper No. RWP19-021. [Online]. Available: <https://ssrn.com/abstract=3403297>
- [5] W. S. Vickrey, "Counterspeculation, auctions, and competitive sealed tenders," *Journal of Finance*, vol. 16, pp. 8–37, 1961. [Online]. Available: <https://api.semanticscholar.org/CorpusID:153426451>
- [6] J. G. Riley and W. F. Samuelson, "Optimal auctions," *The American Economic Review*, vol. 71, no. 3, pp. 381–392, 1981. [Online]. Available: <http://www.jstor.org/stable/1802786>
- [7] V. Krishna, *Auction Theory*, 2nd ed. Burlington, MA: Academic Press/Elsevier, 2010.
- [8] G. d'Eon, N. Newman, and K. Leyton-Brown, "Understanding iterative combinatorial auction designs via multi-agent reinforcement learning," 2024. [Online]. Available: <https://arxiv.org/abs/2402.19420>